Elastic Cuckoo Page Tables

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Outline

Motivation

Elastic Cuckoo Page Tables

Evaluation

Section 1

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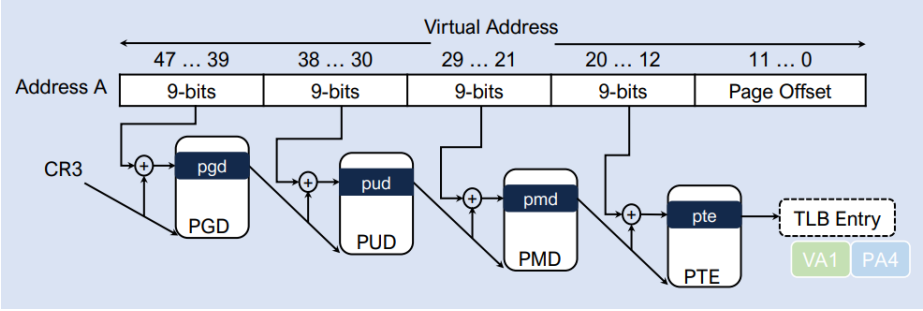
- Page table walks can account for **20-40%** of main memory accesses

Radix Trees



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- **Hash collisions**

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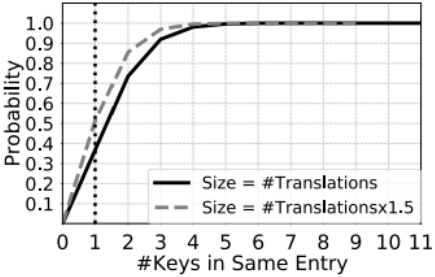
- Collisions lead to more memory accesses as the system walks the collision chain



Section 2

Elastic Cuckoo Page Tables

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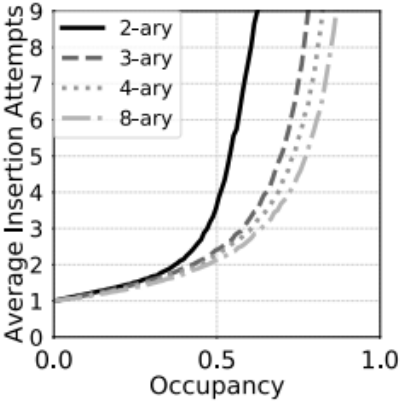
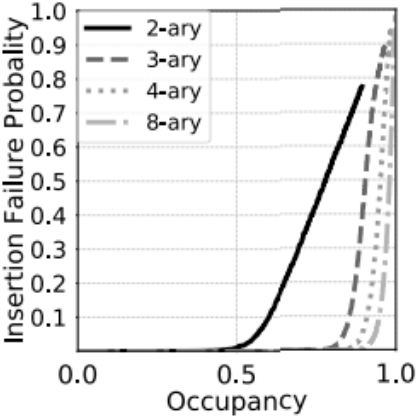
- Finding an element in an $n - ary$ hash table requires only n **independent** lookups

Eviction Overhead





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  - Resizing will be necessary as insertions occur
- **Resizing cuckoo-hashed page tables is expensive**
  - Naive solution requires accessing n ways of old *and* new tables $\rightarrow$ $2 \times n$ memory accesses *per entry*

Paper Contributions

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2. Builds per process **Elastic Cuckoo Page Tables**

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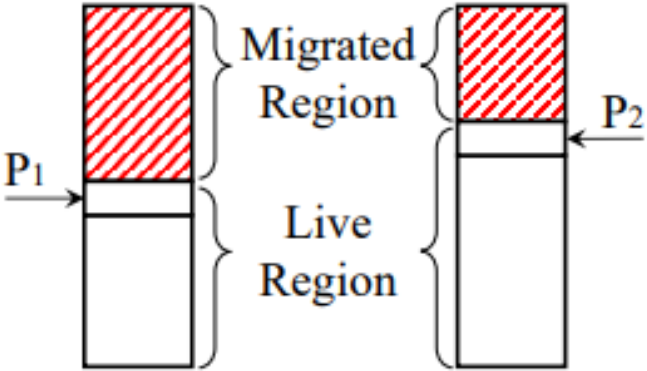
  - Use hash value + pointer to determine if it resides in new or old table
- Minimizes cache pollution $\rightarrow$ never fetch entries from already migrated region

Migration



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  - Separate tables for different page sizes (PUD, PMD, PTE)
  - Allows parallel lookups when page size is unknown
- Single hash table entry contains a VPN tag and multiple packed physical page translation entries
- Utilizes **Cuckoo Walk Tables** to further reduce number of parallel lookups

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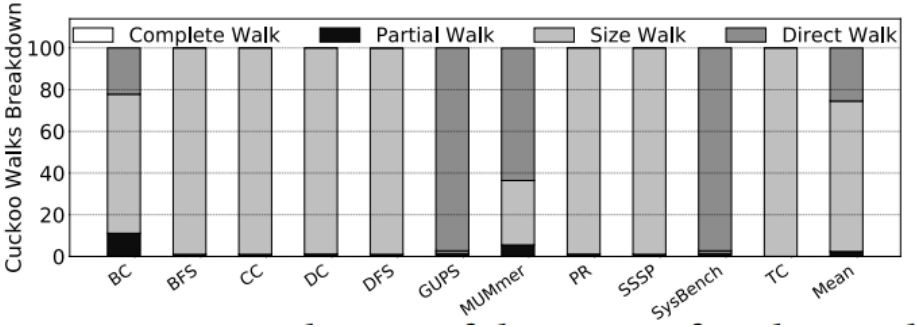
4. **Direct Walk:** CWTs specify size and way of page

Walk Types


Complete Walk
Partial Walk
Size Walk
Direct Walk
Cuckoo Walks Breakdown
100
80
60
40
20
0
BC
BFS
CC
DC
DFS
GUPS
MUMmer
PR
SSSP
SysBench
TC
Mean

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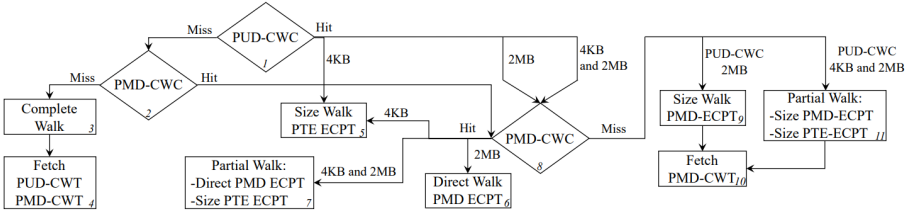
- Cached in MMU (not the PTE-CWT)

ECPT Translation Procedure



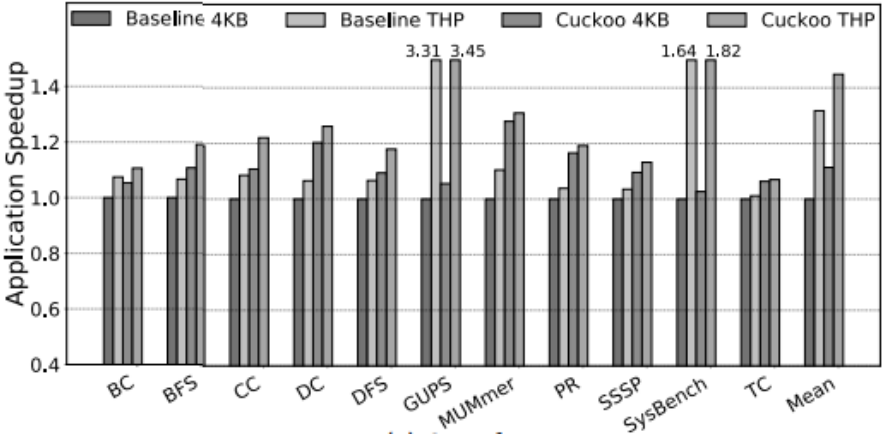
Section 3

Evaluation

Evaluation Methodology

- Full system cycle-level simulation
- Baseline $\rightarrow$ 4-level radix page table architecture (x86-64 like)
- Proposed system evaluated using $3 - ary$ ECPTs
- Benchmark programs run with and without THP

Speedup


Baseline 4KB
Baseline THP
Cuckoo 4KB
Cuckoo THP
Application Speedup
0.4
0.6
0.8
1.0
1.2
1.4
3.31
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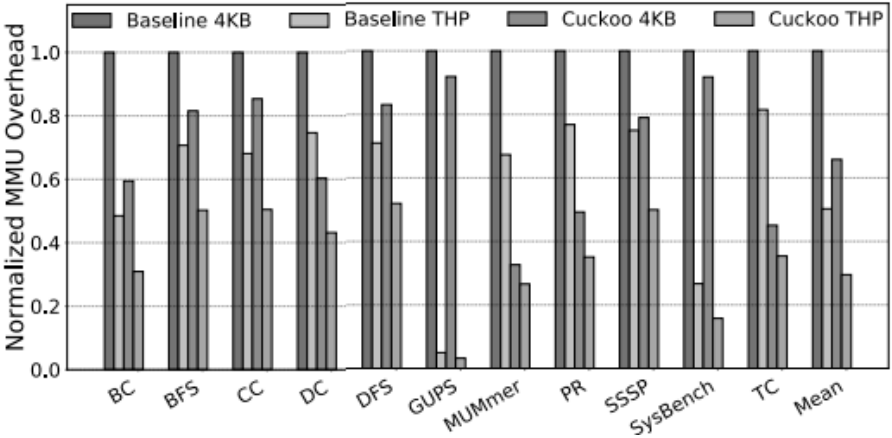
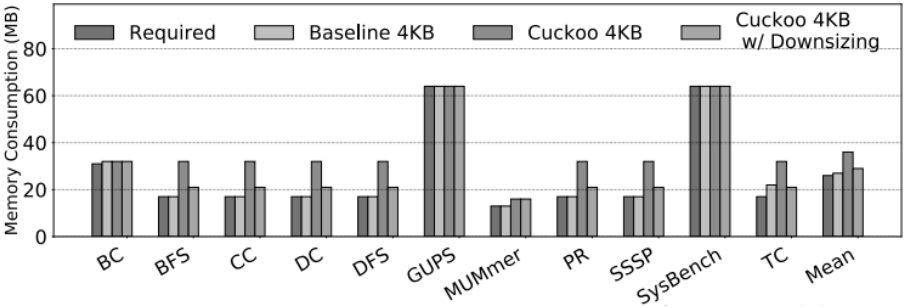
MMU Overhead Reduction



Memory Consumption



The end.